

Arpan Pathak

Seattle, WA | Email: arpan.pathak47@gmail.com | Phone: +1 (206) 306-6059

LinkedIn: [linkedin.com/in/arpan-pathak-272341424](https://www.linkedin.com/in/arpan-pathak-272341424) | GitHub: github.com/arpanpathak | YouTube: [@ArpanPathak](https://youtube.com/@ArpanPathak)

Seattle-based | Local to NVIDIA Redmond (onsite) | Visa: Requires H1B Transfer sponsorship

SUMMARY

Senior Software Engineer, **8 years**: production-grade **AI inference serving**, distributed **backend microservices**, and cloud-native infrastructure at **Amazon, Microsoft, Oracle**. Strong **Python, C++, Rust, CUDA**; **BERT inference at 5,000 QPS** (ONNX, INT8) on **Docker/Kubernetes** with CI/CD. **Passionate about CV & deep learning for road safety** — edge-AI perception on **NVIDIA Jetson Orin** (TensorRT).

NVIDIA GPU HOMELAB

YOLOv8/v11 INT8 → **TensorRT** at **~12 ms** on **NVIDIA Jetson Orin Nano** (Rust, CUDA); real-world 3D scenes, 100% on-device.

- **Deep SORT/Kalman tracking** · Paper: **Intersection Blockage Prediction** | [paper](#)
- Open-source: **Seeing Machines** · **CUDA Kernels** (Rust-to-CUDA) | [github](#)

EXPERIENCE

Software Developer II Microsoft, Redmond, WA Jun 2025 – Aug 2026

- **Containerized Cloud Infrastructure**: zero-trust sovereign-cloud networking with eBPF + Cilium; Kubernetes (AKS) operators reconcile state across regions — container-first, policy-driven systems.
- **Systems Performance**: owned the **C++/C#** → **Rust** migration of Microsoft Information Protection; **~40% lower P99**, **2x throughput**, **3x lower memory**, six-figure savings.
- TechStack: **Rust, C++, Python, Kubernetes (AKS), Docker, CI/CD, Linux**

Senior Software Engineer Oracle Cloud Infrastructure, Seattle, WA Oct 2024 – Mar 2025

- **Backend Services & SDKs**: owned the **Terraform provider** (developer SDK) for Autonomous Database, replacing legacy control-plane tooling; **\$1.2M/month** projected savings.
- TechStack: **Java, Go, Rust, REST/gRPC, Terraform, OCI**

Software Development Engineer II Amazon, Seattle, WA & Hyderabad, India Mar 2021 – Oct 2024

- **Production Inference Serving**: scaled low-latency **BERT-based inference** to **5,000 QPS** (ONNX Runtime, INT8 quantization, latency tuning); validated throughput/latency; cut triage SLA **7 days** → **2 hours**.
- **ML Ranking Pipeline**: owned end-to-end (training → serving) XGBoost + Deep Learning ensemble; **5% ad CTR lift**, **\$5.4M/month** revenue.
- **Event-Driven Microservices**: led a team of 3 building ad-delivery microservices at **<50 ms** via real-time CDC (Kotlin).
- TechStack: **Python, Kotlin, Java, PyTorch, XGBoost, ONNX Runtime, Kafka, AWS**

SDE Razorpay, Bengaluru, India Aug 2020 – Feb 2021

- **Backend Microservices**: P2P Unified Payments Gateway (Go, Protobuf/gRPC): **3M+ transactions/hour** peak, **99.99% uptime**, **150ms P99**.

SKILLS

Languages: Python, C++, Rust, Go, Java, Kotlin, TypeScript, Shell/Bash

AI/ML Inference: ONNX Runtime, TensorRT (INT8), PyTorch, XGBoost, BERT/LLMs, Quantization, MLOps

Backend & Microservices: REST, gRPC, Protobuf, Kafka, CDC, High-Throughput APIs

Containerization & Cloud: Docker, Kubernetes (EKS, AKS), Terraform, Kubebuilder, CI/CD, AWS, Azure

GPU & CV: CUDA, Rust-to-CUDA, GPU Kernels, Jetson Orin, YOLO, 3D Scene Understanding, Edge Inference

EDUCATION

B.Tech, Computer Science & Engineering, RCC Institute of Information Technology, Kolkata | 2018